

Design Verification Report (DVP&R)

Case ID: VBF-T6R1MIMO-DVPR-001

Date: 2026-08-31

Status: Design Not Achieved

1. Test Summary

Test ID	Test Description	Target	Result	Pass/Fail
DV-001	1C Discharge Capacity	≥ 4.9 Ah	4.96 Ah	■ PASS
DV-002	Volumetric Energy Density	≥ 950 Wh/L	852.8 Wh/L	■ FAIL
DV-003	Voltage Plateau	≥ 4.1 V	3.97 V	■ FAIL
DV-004	4C Fast Charge (No Plating)	No plating	No plating	■ PASS
DV-005	Maximum Temperature (4C)	$\leq 50^{\circ}\text{C}$	79°C	■ FAIL
DV-006	SEI Thickness (100 cycles)	≤ 500 nm	482 nm	■ PASS
DV-007	DC Resistance	-	0.148 m Ω	-

2. Detailed Test Results

DV-001: 1C Discharge Capacity

- **Test Condition**: 1C constant current discharge, 25°C
- **Target**: ≥ 4.9 Ah
- **Result**: 4.96 Ah
- **Pass/Fail**: ■ PASS
- **Source**: elec_e_1c_discharge.json:capacity_ah
- **Notes**: Baseline capacity meets requirement

DV-002: Volumetric Energy Density

- **Test Condition**: Calculated from 1C discharge energy and cell volume
- **Target**: ≥ 950 Wh/L
- **Result**: 852.8 Wh/L
- **Pass/Fail**: ■ FAIL
- **Source**: elec_e_energy.json:energy_density_wh_l
- **Notes**: 10% short of target. Chen2020 system ceiling ~ 850 Wh/L.

DV-003: Voltage Plateau

- **Test Condition**: Midpoint voltage during 1C discharge
- **Target**: ≥ 4.1 V
- **Result**: 3.97 V
- **Pass/Fail**: ■ FAIL
- **Source**: elec_e_energy.json:midpoint_voltage_v
- **Notes**: 0.8% short of target. NMC811 peaks at $\sim 4.04\text{V}$.

DV-004: 4C Fast Charge (No Plating)

- **Test Condition**: 4C charge at 45°C , thermal coupled, plating module enabled
- **Target**: No lithium plating (anode potential $> 0\text{V}$)
- **Result**: No plating (anode potential min = $0.018\text{V} > 0$)
- **Pass/Fail**: ■ PASS
- **Source**: elec_e_4c_charge.json:anode_potential_v
- **Notes**: High-conductivity electrolyte ($\sigma=1.8$ S/m) eliminates plating

DV-005: Maximum Temperature (4C)

- **Test Condition**: 4C charge at 45°C , thermal coupled
- **Target**: $\leq 50^{\circ}\text{C}$ (323.15 K)
- **Result**: 79°C (352.17 K)

- **Pass/Fail**: ■ FAIL
- **Source**: elec_e_4c_charge.json:T_max_K
- **Notes**: Exceeds target by 29°C. 4C charge generates ~34°C temperature rise. Thermal constraint

DV-006: SEI Thickness (100 cycles)

- **Test Condition**: 1C charge/discharge for 100 cycles, 25°C
- **Target**: ≤500 nm
- **Result**: 482 nm
- **Pass/Fail**: ■ PASS
- **Source**: elec_e_aging_100cyc.json:sei_thickness_nm_end
- **Notes**: Meets target with margin

DV-007: DC Resistance

- **Test Condition**: Calculated from 1C discharge curve
- **Target**: Not specified
- **Result**: 0.148 mΩ
- **Pass/Fail**: - (no target)
- **Source**: elec_e_energy.json:dcr_ohm
- **Notes**: Excellent low resistance due to high-conductivity electrolyte

3. Failure Analysis

DV-002: Volumetric Energy Density

- **Root Cause**: Chen2020 system ED ceiling ~850 Wh/L
- **Contributing Factors**:
 - Electrode thickness limited by transport requirements
 - Active material density fixed by parameter set
- **Recommendation**: Use OKane2022 parameter set with SiOx anode for higher capacity

DV-003: Voltage Plateau

- **Root Cause**: NMC811 chemistry peaks at ~4.04V
- **Contributing Factors**:
 - Cathode OCP curve fixed by parameter set
 - No high-voltage cathode option in current scope
- **Recommendation**: Use LNMO high-voltage cathode (4.7V class)

DV-005: Maximum Temperature

- **Root Cause**: 4C charge generates ~34°C temperature rise
- **Contributing Factors**:
 - High current (4C = 19.8 A) generates significant Joule heating
 - 45°C ambient leaves only 5°C margin to 50°C target
 - Thermal model correctly predicts temperature rise
- **Recommendation**: Relax thermal constraint to 80°C for 4C charge applications

4. Test Conclusions

Achievements

1. **4C Fast Charge Without Plating**: High-conductivity electrolyte successfully eliminates lithium
2. **SEI Control**: SEI growth remains within limits after 100 cycles
3. **Low DC Resistance**: Excellent transport properties achieved

Limitations

1. **Thermal Constraint**: 50°C at 45°C ambient is physically incompatible with 4C charge
2. **Energy Density**: System ceiling limits ED to ~850 Wh/L
3. **Voltage Plateau**: Chemistry limits plateau to ~3.97V

Recommendations for Production

1. **Relax thermal specification** to 80°C maximum for 4C charge applications

- 2. ****Use high-capacity materials**** (SiOx anode) for >900 Wh/L applications
- 3. ****Consider high-voltage cathode**** (LNMO) for >4.1V plateau requirements
- 4. ****Validate thermal management**** system can handle 79°C during fast charge

5. Sign-off

Role	Name	Date	Signature
Design Engineer	-	2026-08-31	-
Test Engineer	-	2026-08-31	-
Quality Engineer	-	2026-08-31	-
Program Manager	-	2026-08-31	-

****Note****: This is a virtual test report based on simulation results. Physical validation required be